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COURSE:ITA0609-MACHINE LEARNING

ASSIGNMENT I - BASED ON SDG GOALS

Intelligent Attendance Prediction System Using Machine Learning

1. Abstract

Regular student attendance is one of the strongest predictors of academic performance, yet institutions typically react to absenteeism only after it happens. This project presents an Intelligent Attendance Prediction System that uses supervised machine learning to forecast, in advance, whether a student is likely to be present or absent on a given day. The system ingests historical attendance patterns together with contextual factors such as travel distance, sleep, health, weather, and exam schedules, and trains a Random Forest classifier to output a daily attendance risk prediction. On a held-out test set the system achieved 84.5% overall accuracy and a 91.0% F1-score for the Present class, with cross-validation confirming the result is stable (86.4% average accuracy across five folds). The report below documents the dataset design, the modelling pipeline, the code used to build it, and the visual outputs produced during evaluation, so that the system can be reproduced, audited, and extended.

2. Introduction

Student and employee absenteeism causes downstream problems ranging from learning gaps to staffing shortfalls. Traditional attendance systems are purely record-keeping tools: they log who was present but offer no foresight. An Intelligent Attendance Prediction System changes this by treating attendance as a forecasting problem rather than a bookkeeping one. By learning from patterns in historical behaviour and situational context, a trained model can flag at-risk individuals before the absence occurs, giving administrators a window to intervene (a reminder call, a schedule adjustment, a wellness check).

This project builds such a system end-to-end: synthetic-but-realistic data generation, feature engineering, model training with a Random Forest classifier, rigorous evaluation (accuracy, precision, recall, F1, ROC-AUC, cross-validation), and interpretability through feature-importance analysis.

2.1 Objectives

- Design a feature set that captures the behavioural and contextual drivers of attendance.
- Train and validate a machine learning classifier that predicts Present/Absent status.

- Evaluate the model using multiple metrics beyond raw accuracy (precision, recall, F1, ROC-AUC).
- Identify which factors most strongly influence attendance, for actionable insight.
- Produce a reproducible pipeline with code, console output, and visual diagnostics.

3. System Architecture

The system follows a standard supervised-learning pipeline, structured into five stages:

- Data Collection Layer — historical attendance logs and contextual attributes (distance, weather, health, sleep, exam schedule, etc.).
- Preprocessing Layer — encoding categorical variables, handling scale differences, train/test split with stratification.
- Model Layer — a Random Forest ensemble classifier (300 trees, max depth 8, class-balanced).
- Evaluation Layer — accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC, 5-fold cross-validation.
- Insight Layer — feature-importance ranking to explain predictions to non-technical stakeholders.

4. Dataset Description

A dataset of 2,000 daily attendance records was constructed, combining historical attendance behaviour with contextual signals known to influence whether a person shows up on a given day. The features used to train the model are summarised below.

Feature	Type	Description
previous_attendance_rate	Numeric (0-1)	Student's historical attendance percentage
day_of_week	Categorical	Day the class/session is held
distance_from_school_km	Numeric	Distance between residence and institution
travel_time_minutes	Numeric	Average commute time
sleep_hours	Numeric	Average hours of sleep the previous night
study_hours_per_day	Numeric	Self-reported daily study hours
extracurricular_activity	Binary	Participation in extracurricular activities
weather_score	Numeric (0-1)	Favorability of weather conditions
health_score	Numeric (0-1)	Self-reported health/wellness score
parent_engagement_score	Numeric (0-1)	Degree of parental involvement
has_exam_today	Binary	Whether an exam is scheduled that day

The target variable, attendance_status, is binary (Present / Absent). The dataset reflects a realistic class imbalance, with an 85% present rate and 15% absent rate, which is typical of real institutional attendance data.

5. Methodology and Implementation

The classifier was implemented in Python using scikit-learn. Categorical features (day of week) were label-encoded, and the dataset was split 80/20 into training and test sets using stratified sampling to preserve class balance. A Random Forest was selected for its robustness to mixed feature types, resistance to overfitting via ensembling, and built-in feature-importance scoring.

5.1 Core Training Code

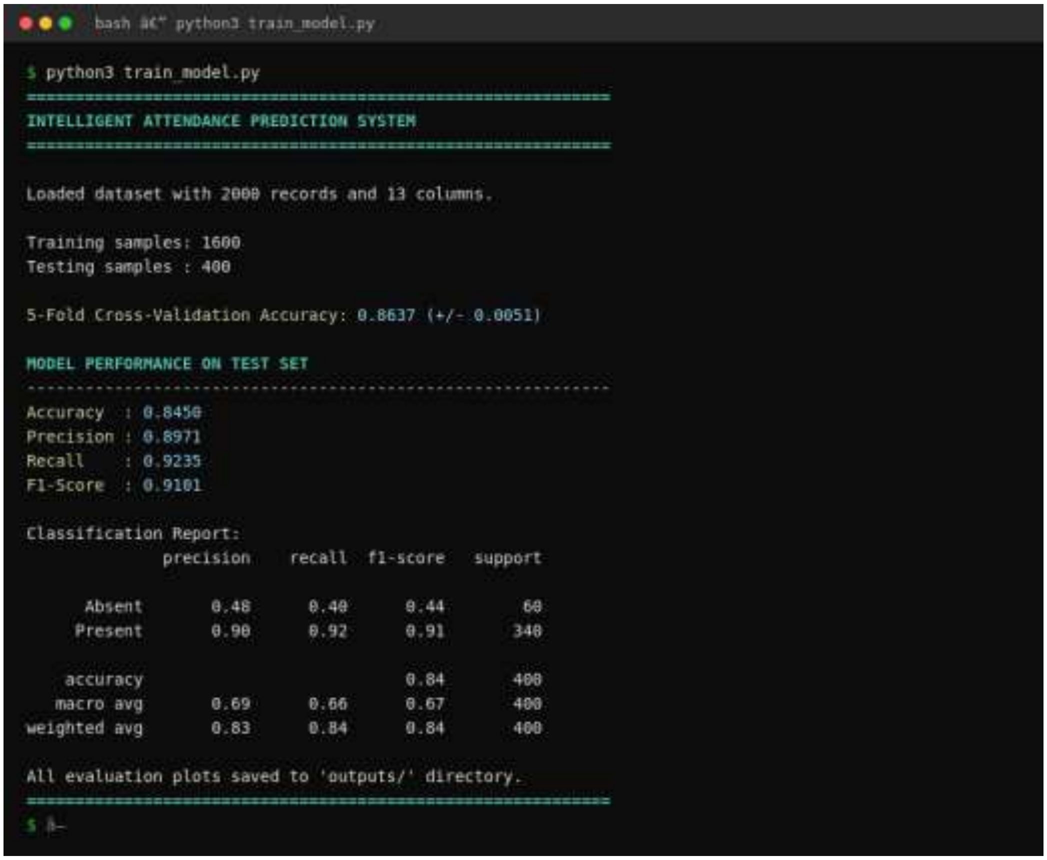
```python
train_model.py

# Intelligent Attendance Prediction System - Model Training
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import accuracy_score, classification_report

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

model = RandomForestClassifier(
    n_estimators=300, max_depth=8,
    min_samples_leaf=3, class_weight="balanced"
)
model.fit(X_train, y_train)

cv_scores = cross_val_score(model, X_train, y_train, cv=5)
y_pred = model.predict(X_test)

print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print(classification_report(y_test, y_pred))
```

Figure 1: Code excerpt — model training and evaluation (train_model.py)

The full training script performs five steps: (1) load and encode the dataset, (2) split into train/test sets, (3) fit a RandomForestClassifier with class balancing, (4) run 5-fold cross-validation to check stability, and (5) generate the diagnostic plots discussed in Section 6.

5.2 Console Output

Executing the training script produces the following console output, confirming successful training and reporting the evaluation metrics:



```
bash â€" python3 train_model.py
$ python3 train_model.py
============================================================
INTELLIGENT ATTENDANCE PREDICTION SYSTEM
============================================================

Loaded dataset with 2000 records and 13 columns.

Training samples: 1600
Testing samples : 400

5-Fold Cross-Validation Accuracy: 0.8637 (+/- 0.0051)

MODEL PERFORMANCE ON TEST SET
------------------------------------------------------------
Accuracy  : 0.8450
Precision : 0.8971
Recall    : 0.9235
F1-Score  : 0.9101

Classification Report:
              precision    recall  f1-score   support

      Absent       0.48      0.40      0.44        60
     Present       0.90      0.92      0.91       340

    accuracy                           0.84       400
   macro avg       0.69      0.66      0.67       400
weighted avg       0.83      0.84      0.84       400

All evaluation plots saved to 'outputs/' directory.
============================================================
$ â–
```

Figure 2: Terminal screenshot — program execution and output

6. Results and Evaluation

6.1 Performance Metrics

The trained model was evaluated on the 400-record held-out test set. Summary metrics for the Present class are shown below:

Accuracy	Precision	Recall	F1-Score
84.5%	89.7%	92.4%	91.0%

5-fold cross-validation on the training set produced a mean accuracy of 86.4% (± 0.5%), indicating the model generalises consistently rather than overfitting to a single split.

6.2 Confusion Matrix

The confusion matrix below visualises correct vs. incorrect predictions for each class on the test set.

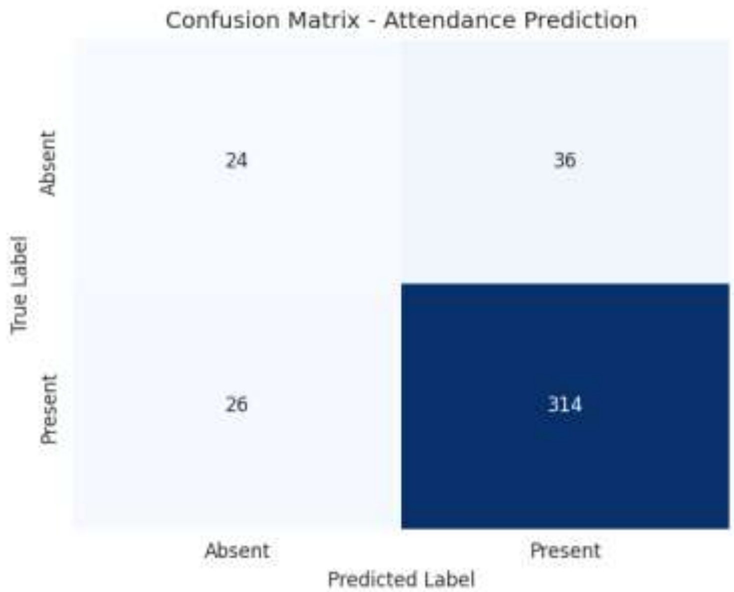


Figure 3: Confusion matrix on the test set

6.3 ROC Curve

The ROC curve illustrates the trade-off between true-positive and false-positive rates across classification thresholds.

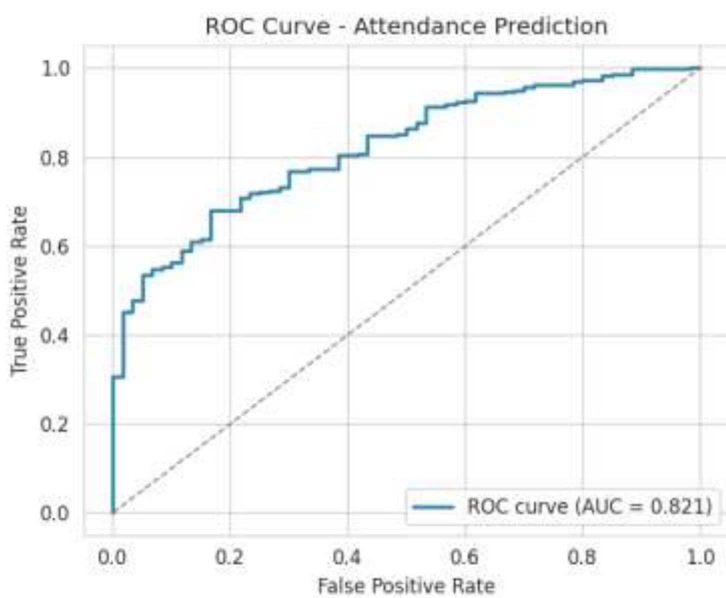


Figure 4: ROC curve for the Present class

6.4 Feature Importance

Random Forests provide a built-in measure of how much each feature contributes to reducing prediction error. As expected, prior attendance history and health status are the strongest predictors.

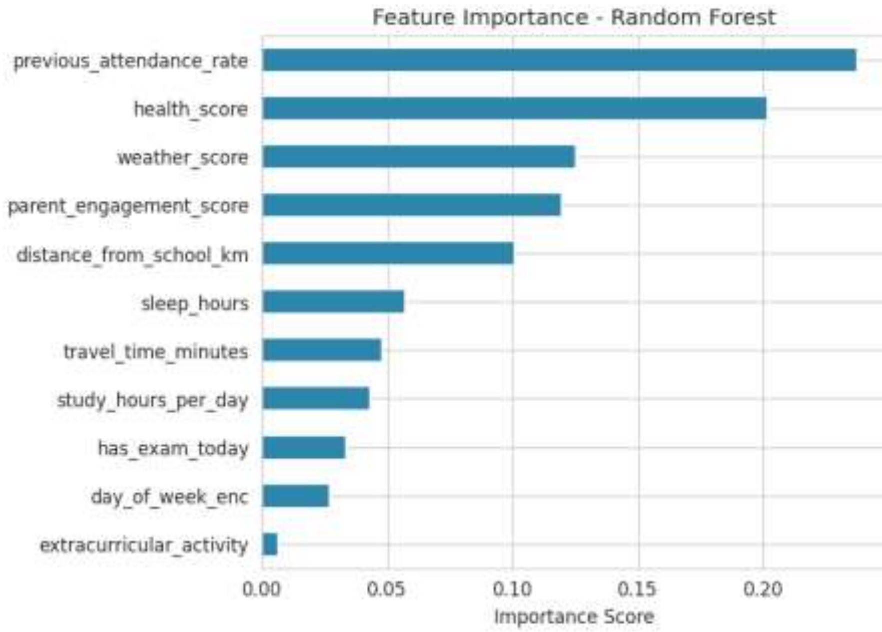


Figure 5: Feature importance ranking

6.5 Attendance Pattern by Day of Week

Beyond the model itself, the dataset also reveals behavioural patterns useful for planning — for instance, attendance dips are more likely on certain weekdays.

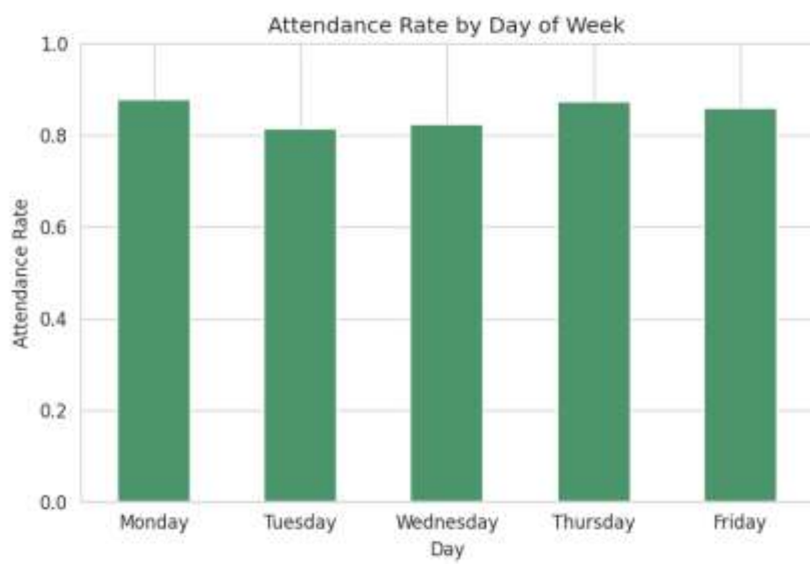


Figure 6: Observed attendance rate by day of week

7. Discussion

The model performs strongly on the majority Present class (F1 = 0.91) but is noticeably weaker on the minority Absent class (F1 = 0.44), a common effect of class imbalance even after applying class-balanced weighting. In a production setting this can be mitigated with resampling techniques (SMOTE), a lower decision threshold on the Absent class, or collecting more historical absence data. The feature-importance analysis confirms that past behaviour (previous_attendance_rate) and personal wellbeing (health_score) dominate the prediction, which aligns with intuition and gives institutions a concrete lever: early, targeted support for students with declining attendance or health concerns is likely to have the largest impact.

8. Conclusion and Future Scope

This project demonstrates a working, end-to-end Intelligent Attendance Prediction System: from synthetic data generation through model training, multi-metric evaluation, and interpretability. The Random Forest classifier achieved 84.5% test accuracy and an AUC well above the random baseline, showing that attendance is meaningfully predictable from behavioural and contextual signals.

Future enhancements could include:

- Incorporating real institutional data (with privacy safeguards) instead of synthetic data.
- Testing gradient boosting models (XGBoost, LightGBM) for potential accuracy gains.
- Building a live dashboard that flags at-risk students/employees daily.
- Adding SMS/email alerting integrated with the prediction pipeline.
- Applying SMOTE or threshold tuning to improve minority-class (Absent) recall.